

Synopsis: Early-Universe Specialization of Hysterical Instability

Instability inheritance, rapid-formation reading, and the FLRW timing suite

Andrew Bruce Boyd

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Computational note. Proofs, exploratory experiments, and paper writing were assisted by generative AI. Mathematical theorems stated in this paper are formally verified in Lean 4 in the companion specifications, as faithful ports of the TeX arguments at the abstractions recorded there, except results explicitly tagged as *literature hypotheses* (standard theorems cited from the literature and not re-proved here) or *physical inputs* (modeling assumptions outside mathematics).

Abstract

Short guide to the early-universe specialization note. Instability and threshold times are inherited from the response-theory paper. What is new is the cosmological closure, a dust-rate comparison, an interpretive reading as rapid early galaxy formation, and the full FLRW timing suite formerly developed as a follow-on numerical paper.

1 Why this note exists

The response-theory paper already proves closed-loop instability when $\alpha(J) > 0$, and finite logarithmic threshold times for nonzero unstable projections. This note checks that a minimal post-decoupling cosmological model lands in that supercritical phase, then asks whether the same loop can act fast enough to matter for the earliest galaxies.

2 Cosmological closure

Local reservoir linearization $\dot{q} + \gamma q = \sigma \delta$ with attractive dust closure produces a frozen Jacobian with

$$P(\lambda) = (\lambda + \gamma)(\lambda^2 + 2H\lambda - A) - A\beta\sigma.$$

Positive coupling gives $P(0) < 0$, hence $\alpha(J) > 0$, so the inherited instability package applies. A response-coupled root also satisfies $\lambda_+ > \lambda_b$ for ordinary dust.

3 Interpretation

Candidate mechanism for rapid early galaxy formation: baryonic seeds amplify a response sector that feeds back into gravity, without a CDM perturbative scaffold. JWST has spectroscopically confirmed luminous galaxies at $z \simeq 14$.

4 FLRW timing suite

Planck-like background; $x = \beta q$; $\kappa = \beta\sigma/H$, $g = \gamma/H$ held constant; baryon-only clustering; fiducial start at $z_{\text{on}} = 1089$ with $\delta_i = 10^{-5}$.

Proposition 4.1 (Timing viability at $z \simeq 14$). *For $g = 0$, $\kappa_{\text{crit}}(z_{\text{nl}} = 14) \simeq 129.8$.*

Further numerical results retained from the timing suite:

- Loss raises κ_{crit} ($\simeq 142, 171, 254$ for $g = 0.3, 1, 3$).
- Delayed activation raises it more strongly ($\simeq 197\text{--}540$ for $z_{\text{on}} = 600\text{--}200$).
- Near threshold, $x = \beta q$ amplifies ahead of δ .
- Smaller seeds cross later at fixed κ (logarithmic timing, not weaker exponent).

This is timing possibility, not an abundance fit. CMB transfer, stellar masses, and luminosity functions remain open.